

Motor Imagery Brain Functional Connectivity Analysis via Coherence

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Abstract— Performance of motor imagery (MI)-based brain computer interfaces (BCIs) highly depends on the extraction of features from different brain neural processes. Commonly designed BCIs use band power changes in single channel electroencephalograms (EEGs) to discriminate different MI tasks. In this paper, we studied the information about interactions of spatially separated brain areas by considering the relationships between brain source signals through functional connectivity. We investigated dynamic time-frequency connectivity patterns during four motor imagery movements including left hand, right hand, both feet and tongue by means of Coherence measure estimated from multivariate adaptive autoregressive model coefficients. To tackle the volume conduction problem, sensor space signals were transformed into source space using independent component analysis technique. We showed distinct time-varying Coherence for different motor imageries. For tongue MI movement, bilateral connectivity was observed in hand areas of both hemispheres. The similar connectivity was detected for feet MI task, but with more focus on right hemisphere. During right and left hand MI, bilateral and contralateral brain connectivities were observed respectively. Results provided valuable information for possible classification of MI tasks by considering brain source functional connectivity patterns for BCIs.

Keywords—motor imagery, connectivity, coherence, brain computer interface

I. INTRODUCTION

A brain-computer interface (BCI) is a communication device which reads brain activities and transforms them to control commands for devices such as a wheelchair or a neuroprosthesis [1]. Common BCI systems rely on electroencephalogram (EEG) as a non-invasive, inexpensive, lightweight and portable neuroimaging technique. The most commonly used experimental paradigm in this context is motor imagery (MI). During the imagination of movements of body parts such as the hand or feet, sensorimotor rhythms (SMRs) generated from the somatosensory and motor cortices modulate their oscillations depending on the body part being imagined. Typical signal processing steps of EEG-based BCIs consist of preprocessing, feature extraction and classification. Previous researches have reported that when the subject imagines limb movement, specific frequency components of EEG such as the mu and central beta rhythms are (de)synchronized over

the contralateral (ipsilateral) sensorimotor area [2]. Early studies on MI movement pattern discrimination were based on quantification of event-related desynchronization (ERD) and event-related synchronization (ERS) using band power. Overall accuracy of MI-based BCIs was highly dependent on the extraction of informative feature patterns in ERD/ERS corresponding to specific body parts. However, the feature extraction based on ERD/ERS patterns requires time-consuming and complex analysis in every EEG channel. Moreover, it has been shown that the MI responsive frequency bands are highly variable for inter- and intra-subjects [3] which indicates the instability of such BCIs. Besides, these BCIs are spatially limited in EEG recording locations on the somatosensory and motor cortices. However, several EEG studies reported that MI activates primary sensorimotor areas [4]. Furthermore, it has been revealed that during the performance of cognitive tasks, many different parts of the brain are activated and communicate with each other which make it difficult to isolate one or two areas where the activity takes place [5]. For instance, the supplementary motor area (SMA), prefrontal area, premotor cortex, cerebellum and basal ganglia are activated during MI movements [6]. Accordingly, new types of features that contain information about relationships between channels have been proposed. The interaction of spatially separated brain areas can be observed through functional or effective connectivity. Functional connectivity reveals statistical dependencies between different separated areas of information processing neuronal populations. Several studies have investigated functional connectivity during MI movements in BCI context, [7-9]. These studies estimated connectivity measures through bivariate routine. However, Kus et al., (2004) reported that bivariate methods either based on phases of bivariate coherences or bivariate Granger causality measure lead to inaccurate results [10]. It was proved that multivariate measures derived from multivariate autoregressive (MVAR) modeling of multichannel EEG signals is a superior method to estimate connectivity measures [10]. Moreover, Spiegler et al., (2004) analyzed brain connectivity in static framework which is a premature assumption and delivers misleading results.

EEG recordings are spatially diffused while traversing from the source regions to the sensors by a process called volume conduction. Due to volume conduction, a signal generated by a source affects different sensors on the scalp surface; therefore,

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the signal recorded from an EEG sensor represents the superposition of the activity of several active sources throughout the brain. This delivers spurious zero-lag connectivity between scalp EEG signals and can cause misinterpretation of sensor-space EEG analysis [11]. To address this problem, various solutions have been proposed: connectivity measures that suppress instantaneous effects [12], inclusion of an instantaneous term in VAR models [13-14], and modeling the EEG as a mixture of sources [15]. Among these the latter have been widely used in BCI context.

Recently, hand and foot MI movements were classified by applying different connectivity measures derived from MVAR model at source level [15]. In this paper we follow the source-space MVAR approach for estimating the dynamic functional connectivity during four motor imagery movements. Instead of classification, we study time-frequency dynamics of brain source interactions quantified by coherence measure. Our approach includes blind source separation of EEG brain sources using independent component analysis, estimating dynamic multivariate functional connectivity using coherence derived from MVAR model and representing time-frequency connectivity maps of statistical significant coherence among connected EEG sources for each of four different motor imageries, left hand, right hand, both feet and tongue. The main goal is to study dynamic time-frequency connectivity pattern unique to each MI. This may provide new information about whole-brain connectivity differences among the MI tasks in BCI context.

II. METHODS

A. Independent Component Analysis (ICA)

ICA is a powerful technique for separating artifacts from sensor-space EEGs and related data as well as separating brain source activities. It distinguishes signals in recorded multi-channel data mixtures whose time courses are maximally independent of each other and in this sense contribute maximally distinct information to the recorded data. A very well-known example application for identifying source signals is “cocktail party problem”. Imagine that we are in a room where two people are speaking and another one is singing simultaneously. If three microphones are sited in three different places in the room, they record a particular linear combination of the three voices denoted by $x_1(t)$, $x_2(t)$ and $x_3(t)$. Each of these measured signals is a weighted sum of the signals emitted by the three speakers referred by $s_1(t)$, $s_2(t)$ and $s_3(t)$. This can be expressed as the following equation

$$\begin{aligned} x_1(t) &= a_{11}s_1 + a_{12}s_2 + a_{13}s_3 \\ x_2(t) &= a_{21}s_1 + a_{22}s_2 + a_{23}s_3 \\ x_3(t) &= a_{31}s_1 + a_{32}s_2 + a_{33}s_3 \end{aligned} \quad (1)$$

This can be summarized as

$$X = AS \quad (2)$$

If the parameters a_{ij} are known, the linear equation (1) can be solved and the source signals can be obtained. ICA can be applied to estimate the a_{ij} based on their independence information and separate the three source signals. In fact, ICA decomposes a multivariate signal into independent non-gaussian signals.

Recorded scalp EEG data are a mixture of some underlying components of brain activity which is very similar to the cocktail-party problem. So, when using ICA to solve the equation (2) for analysis of EEG signals, A is the data matrix of size (number of channels by number of time points), W is an unmixing matrix of size (number of ICA components by number of channels), and S is the ICA component activation time courses of size (number of ICA components by number of time points).

B. MVAR Model for Connectivity Estimation

After using ICA, instantaneous dependencies between signals are minimized and delayed dependencies are preserved which can be modelled with MVAR model. For $\mathbf{y}_n = (y_{n1}, \dots, y_{nd})^T$, $n=1, \dots, N$ EEG signals of length N of d sources and a TV-MVAR model of order p can be defined as

$$\mathbf{y}_n = \sum_{k=1}^p \mathbf{A}_k \mathbf{y}_{n-k} + \mathbf{v}_n \quad (3)$$

where $\{\mathbf{A}_k\}_{k=1}^p$ are $d \times d$ matrices of TV-MAR coefficients at time lag k and $\mathbf{v}_n$ is a $d \times 1$ residual at time n . By transforming the model equation (3) into the frequency domain, a spectral representation of the model is as

$$\mathbf{v}(z) = \left(\mathbf{I} - \sum_{k=1}^p \mathbf{A}_k z^{-k} \right) \mathbf{y}(z) \quad (4)$$

We define the $[M \times M]$ matrix function $X(z)$, which is the inverse of the multivariate transfer function $H(z)$:

$$X(z) = \mathbf{I} - \sum_{k=1}^p \mathbf{A}_k z^{-k} \quad (5)$$

$$\mathbf{y}(z) = X(z)^{-1} \mathbf{v}(z) = H(z) \mathbf{v}(z) \quad (6)$$

Now functional connectivity Coherence from MVAR model can be computed. The cross-spectral density (S) is obtained by

$$S(z) = H(z) \Sigma_x H(z)^H \quad (7)$$

The superscript indicates the Hermitian operator (transposed complex conjugate of matrix H). Then, coherence can be estimated via normalizing S by the corresponding auto-spectra:

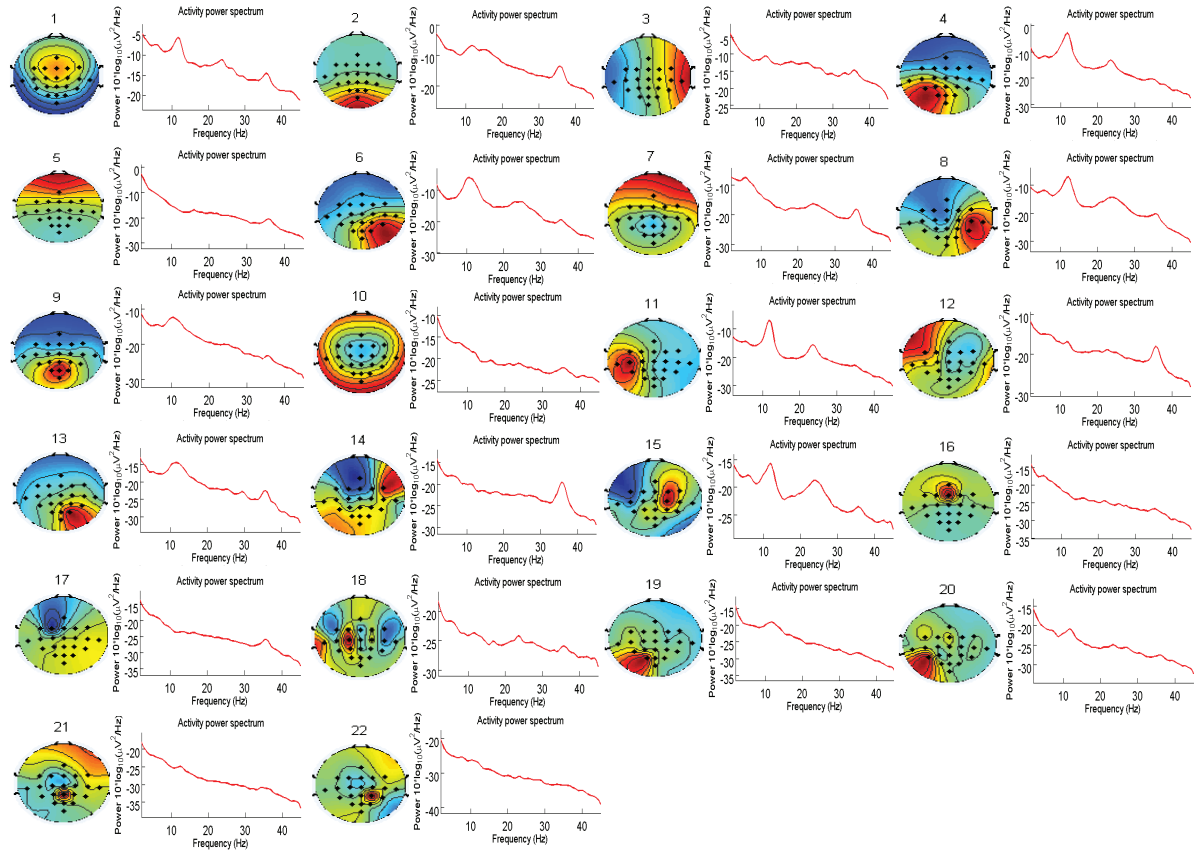


Fig. 1. The component scalp maps and their power spectrum activities.

$$Coh_{ij}(z) = \frac{|S_{ij}(z)|}{\sqrt{S_{ii}(z)S_{jj}(z)}} \quad (8)$$

III. EXPERIMENTAL RESULTS

A subset of the BCI competition IV data set 2a with four MI tasks, left hand, right hand, both feet and tongue from one subject was used for this study. The subject was asked to perform imagery movement which was started according to a cue from $t=3s$ until $t=7s$ while EEG data was recorded with 22 channels at sampling frequency 250Hz. The analysis was carried out in EEGLAB with the SIFT toolbox [16]. In the first step, scalp recorded signals were band-passed filtered between 1-50 Hz to envelope the interested frequencies. Next, common average reference was used to re-reference the EEG signals as a better choice than other referencing methods such as linked mastoid reference [11]. After that, data was epoched and all single-trials were baseline corrected by subtracting the mean of the samples before cue-onset. Then, artifactual epochs were removed and to obtain source signals, extended Infomax ICA [17] was employed. Good neural components related to neural process in the brain were selected for further processing by investigating the components topographical maps and power spectrum activities.

Figure 1 shows the component scalp maps and power spectrum activities of components. According to this figure, component 3 was a typical muscle activity artifact as it was spatially localized and showed high power at high frequencies. Component 5 was a typical eye blink artifact since its scalp map showed a strong far-frontal projection and its EEG power spectrum decreased smoothly. Components 12, 14 and 20 did not show clear brain activity as they were not properly covered due to insufficient number of electrodes in these areas. Besides, there were no considerable spectral peaks at typical EEG frequency for these components. Components 16, 17, 18, 21 and 22 were spatially localized and accounted as electrode noise, typical artifact at the scalp electrode. Hence, 12 components 1, 2, 4, 6, 7, 8, 9, 10, 11, 13, 15 and 19 were selected as brain neural activity for further process and were renumbered accordingly from 1 to 12.

The selected components were then fit into a MVAR model using Vieira-Morf algorithm. The most optimum model was formed after trying different values and finally window length equal to 0.3 with 0.05 overlap were chosen. Moreover, the optimum model order p was selected using Hannan-Quinn criterion (HQ), $p=10$. Finally, functional connectivity Coherence was computed using the fitted model coefficients. Figure 2 illustrates the most significant values of inter-source Coherence of each MI task at 99% level of significance after applying the surrogate data method. We expected to observe

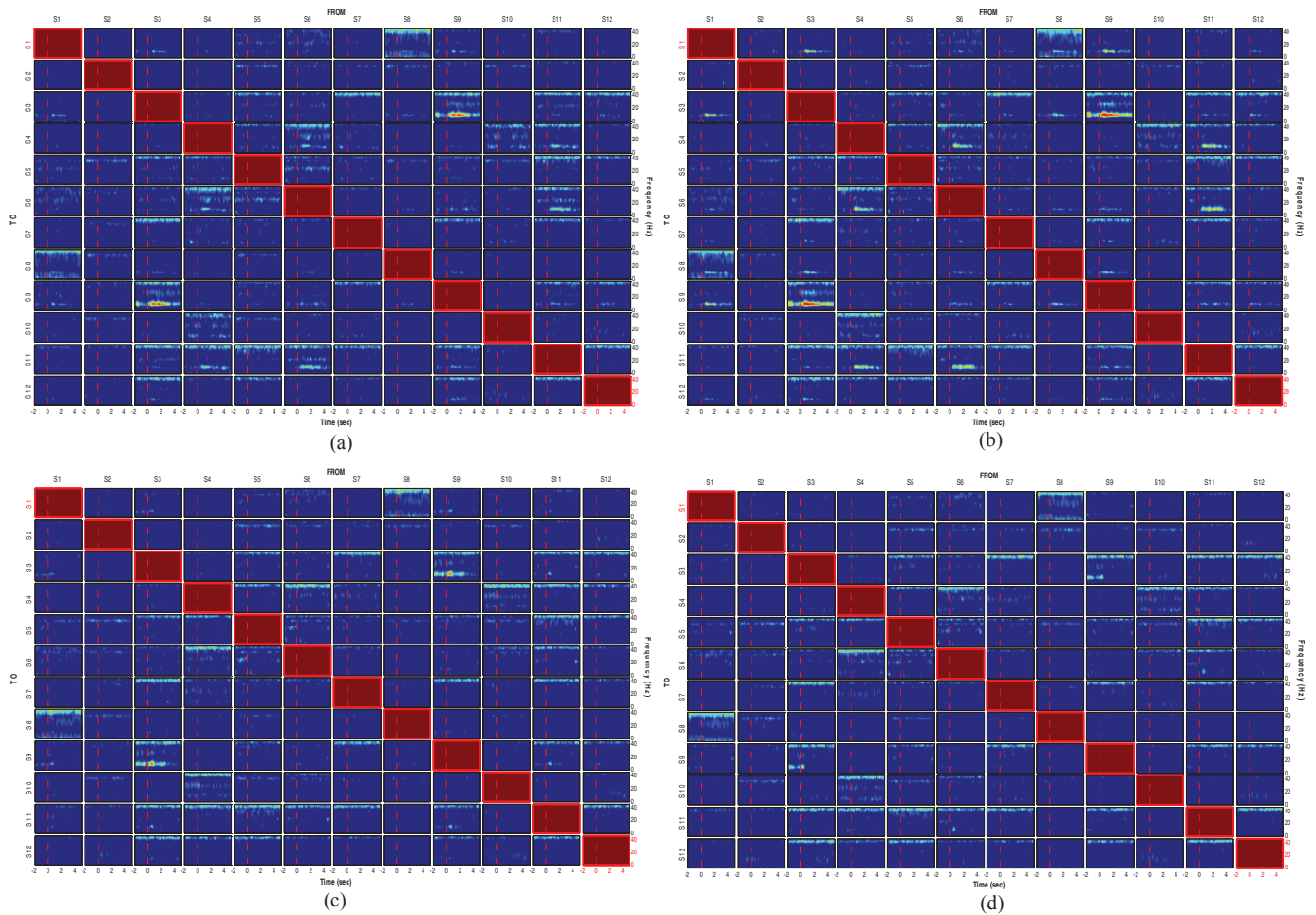


Fig. 2. Coherence functional connectivity during the imagination of (a) Feet (b) Tongue (c) Left hand (d) Right hand movements. S1-S12 indicates components 1-12.

the increase of Coherence according to each MI after cue onset, between 1s to 4s. As shown in figure 2(a), for feet MI, there were strong and weak connectivity between two sources in left frontal area within precentral gyrus region (S3-S9) in alpha and beta bands respectively. Moreover, there were several connectivities between sources in parietal and frontal areas: S4-S11 and S9-S11 in alpha band, S5-S6 in beta band, S4-S6, S6-S11 and S9-S10 in alpha as well as beta band and S2-S5 in gamma band. Figure 2(b) indicates that for Tongue MI, there were strong and weak connectivity between sources S3 and S9 in alpha and beta bands respectively. Furthermore, there were several interactions for this MI task in alpha band: S1-S3, S1-S8, S1-S9, S3-S8, S3-S12, S4-S6, S4-S11, S6-S8, S6-S9, S6-S11, S8-S9 and S9-S11. For Left hand MI (Figure 2(c)), the following couplings are obtained: S3-S9 in alpha band, S1-S8 in beta band and S2-S5, S5-S6, S5-S10, S5-S11 and S6-S10 in gamma band. As shown in Figure 2(d), several interactions are formed during Right hand MI: S4-S11 in alpha and beta bands, S1-S8 in beta band, S2-S5, S2-S8, S2-S10, S5-S6, S5-S8 and S5-S9 in gamma band.

These results showed that feet and tongue MI tasks partly shared similar brain network connectivity. Instances are the interactions within left frontal lobe (S3-S9), within right frontal lobe (S6-S11), between right parietal and frontal areas (S4-

S11) and between left and right frontal lobes (S9-S11). To some extent similar brain connectome was also observed for Left and Right hand MI tasks such as the gamma band inter-hemispheric coupling (S2-S8). Figure 2 shows that there was strong common alpha band coherence between S3 and S9 for tongue, left hand and feet MIs although it lasted longer for tongue MI and it was very short for left hand MI.

Regardless of common interactions, there were specific connectivity for each MI task. In particular, for Feet MI, several beta band couplings were detected, i.e. between right parietal and frontal lobes (S4-S6) and within right frontal area (S5-S6; S6-S11). For Tongue MI, an inter-hemispheric source (S8) was connected to left (S3,S9) and right (S6) frontal lobes in alpha band. For Left hand MI, unique connections were found in gamma band between right somatosensory association cortex and right supplementary motor area (S5-S10) as well as right premotor cortex (S6-S10). This emphasises the involvement of contralateral hemisphere during such task. During Right hand MI, an inter-hemispheric parietal source (S2) interacted with right somatosensory association cortex (S10). Moreover, right supplementary motor area was connected to left precentral gyrus (S5-S9). Figure 2 also illustrates that some gamma band connectivity (around 40Hz) might not be related to motor task as they spread throughout

the whole epoch (-2 to 5 sec) and were almost constant over time.

IV. CONCLUSION

In this study, we evaluated the source space functional connectivity measured by Coherence during motor imagery movements of the left hand, right hand, both feet and tongue in one subject. We observed distinct Coherence with respect to different motor imageries. While in gamma band the information was transmitted between right and left hemispheres (bilateral) during Right hand MI, contralateral brain connectivity was seen for left hand MI. Investigation on tongue motor imagery movement showed alpha band bilateral connectivity in hand areas in both hemispheres which was in line with the results in [7]. For feet MI task, connectivity was also observed in hand areas in both hemispheres with more focus on right hemisphere.

Although there is high potential for further improvement, we found some discriminative connectivity pattern for classification of different MI tasks with Coherence for BCIs. In future, extensive dataset including several subjects will be investigated. Datasets with more recording channels will be considered. Other functional and effective connectivity measures will be examined. Source localization will be studied to find dipoles locations corresponding to each selected component.

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